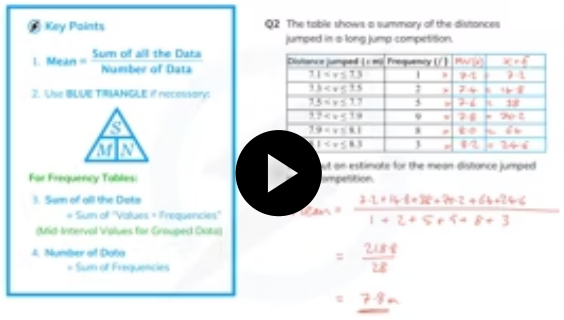
Averages from Grouped Data

Exam code: 0607

Download PDF

Written by: Roger B
Reviewed by: Dan Finlay
Updated on 21 January 2026



Did this video help you? Yes No

Averages from grouped data

What is grouped data?

- Data can be collected into **groups** or **class intervals**
  - It is useful for organising data if you have a lot of individual data points
  - You can present grouped data in a **grouped frequency table**
- Grouped data may be **discrete** or **continuous**
  - Discrete data is numerical data that can only take on specific values, it needs to be counted
    - E.g. Shoe size
  - Continuous data can take **any value** within a range of infinite values, it needs to be **measured**
    - E.g. Length of a foot in cm

Why can I only find an estimate for the mean from grouped data?

- It is impossible to find the **mean** for grouped data, because we don't have access to the original data values
  - i.e. there is no way to find the exact sum of all the data values

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- To do this we use the class **midpoints** as our data values
  - e.g. if a class interval is $150 \le x < 160$
  - we assume that all the data values are equal to the midpoint, 155

Examiner Tips and Tricks

When presented with data in a table it may not be obvious whether the data is grouped or not.

When you see the phrase "**estimate** the mean" you know that you are in the world of grouped data!

How do I find an estimate for the mean from grouped data?

- To find an estimate for the mean from grouped data, complete the following steps:
- **STEP 1**
  Draw an extra **two** columns on the end of a table of the grouped data
  - In the first new column write down the **midpoint** of each **class interval**
  - If the midpoint isn't obvious, add the endpoints and divide by 2
    - e.g. if a class interval is $150 \le x < 160$
    - the midpoint is $\frac{150+160}{2} = \frac{310}{2} = 155$
- **STEP 2**
  Calculate **"frequency" × "midpoint"** (this is often called *fx*)
  - Write these values in the second column you added to the table
- **STEP 3**
  Find the **total** for the ***fx* column**
  - If the question does not tell you the total number of data values (i.e. the total frequency), find the total of the frequency column also
- **STEP 4**
  Estimate the **mean** by using the **formula**
  - $\text{estimated mean} = \frac{\text{total of (midpoints} \times \text{frequencies)}}{\text{total frequency}}$
  - i.e. **divide** the total of the *fx* column by the total number of data values

How do I find the modal class?

- For grouped data we talk about the **modal class** instead of the mode
  - This is the class with the **highest frequency**
- Find the highest frequency in the table

How do I find the class interval that the median lies in?

- Find the **position** of the median using $\frac{n+1}{2}$, where n is the number of data values (total of the frequency column)
- Use the table to deduce the **class interval** containing the $\left(\frac{n+1}{2}\right)^{\text{th}}$ value
  - e.g. if the median is the 7th value and the frequency of the first two class intervals are 4 and 7
    - the median will lie in the second class interval of the table
- Note that rather than 'the median' we refer to the '**class interval containing the median**'

Examiner Tips and Tricks

Be careful not to confuse the **modal class or interval containing the median** with its **frequency**.

e.g. if the highest frequency in the table is 34, corresponding to the class interval $40 \le x < 50$ then the modal class is $40 \le x < 50$, not '34'.

This is a common mistake in exams!

Worked Example

The weights of 20 three-week-old Labrador puppies were recorded at a vet's clinic. The results are shown in the table below.

Weight, w kg	Frequency
$3 \le w < 3.5$	2
$3.5 \le w < 4$	4
$4 \le w < 4.5$	6
$4.5 \le w < 5$	5
$5 \le w < 5.5$	2
$5.5 \le w < 6$	1

(a) Estimate the mean weight of these puppies.

Answer:

First add two columns to the table
Complete the first new column with the midpoints of the class intervals

Weight, w kg	Frequency	Midpoint	"fx"
$3 \le w < 3.5$	2	**3.25**	**2 × 3.25 = 6.5**
$3.5 \le w < 4$	4	**3.75**	**4 × 3.75 = 15**
$4 \le w < 4.5$	6	**4.25**	**6 × 4.25 = 25.5**
$4.5 \le w < 5$	5	**4.75**	**5 × 4.75 = 23.75**
$5 \le w < 5.5$	2	**5.25**	**2 × 5.5 = 10.5**
$5.5 \le w < 6$	1	**5.75**	**1 × 5.75 = 5.75**
Total	**20**		**87**

Now we can find the mean using

$$\text{estimated mean} = \frac{\text{total of (midpoints} \times \text{frequencies)}}{\text{total frequency}}$$

$$\text{estimated mean} = \frac{87}{20} = 4.35$$

4.35 kg

(b) Write down the modal class.

Answer:

The highest frequency in the table is 6
This corresponds to the interval $4 \le w < 4.5$

$4 \le w < 4.5$

(c) Find the interval that contains the median.

Answer:

There are 20 dogs
The median interval will be the interval containing the 10.5^{th} dog
Keep a running total

Weight, w kg	Frequency	Running Total
$3 \le w < 3.5$	3	3
$3.5 \le w < 4$	4	3 + 4 = 7
$4 \le w < 4.5$	6	7 + 6 = 13
$4.5 \le w < 5$	5	13 + 5 = 18
$5 \le w < 6$	2	18 + 2 = 20

The 10.5^{th} dog is in the third interval

The median is in the interval $4 \le w < 4.5$

Teacher only

Test your students on this in just a few clicks

Quizzes or homework in minutes

Try Test Builder

Test yourself

Was this revision note helpful? Yes No

Previous Next

Course Study tools All topics

Resources

Learning Hub
Scholarship
Join
Past Papers
Solution Banks
Sitemap

Dashboard

Overview
Brand ambassador
Profile
Subscription
Settings
Log out

Company

About us
Exam Specificity
Content Quality
Promotions
Jobs
Terms
Privacy
Cookie Policy
Help and Support

Subjects

Biology
Chemistry
Physics
Maths
Geography
English Literature
Psychology
All Subjects